

**PROPOSED MARKING SCHEME**

**A-LEVEL BIOLOGY SEMINAR**

**KIBULI S. S.**

**ON 17-09-2022**

**CONTACT:**

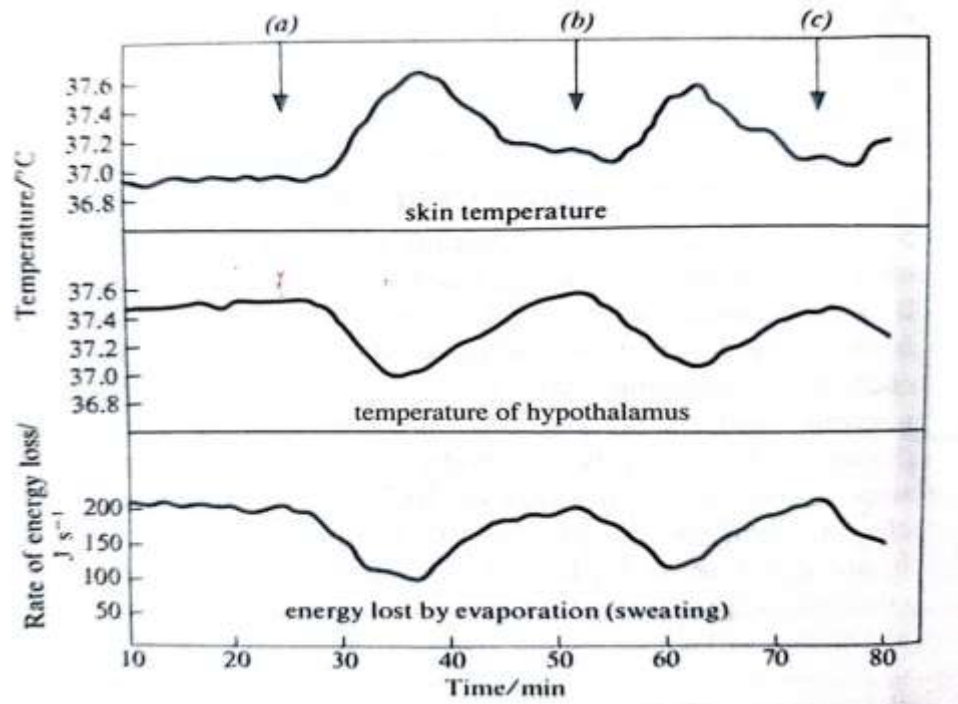
**0757438443**

**0772432505**

**Email:**

**[Ssembogga6236@gmail.com](mailto:Ssembogga6236@gmail.com)**

- The graphs below show the relationship between skin temperature, temperature of the hypothalamus and the rate of evaporation for a human in a warm chamber maintained at  $45^{\circ}\text{C}$ . Iced water was swallowed at points labelled (a), (b) and (c).



- Describe the relationship between the temperature of the hypothalamus and the rate of sweating.**

Before taking iced water, as the temperature of the hypothalamus remains almost constant, the rate of sweating remains almost constant; After taking iced water, as the temperature of the hypothalamus decreases, the rate of sweating decreases; As the temperature of the hypothalamus increases, the rate of sweating increases.

- Between the first two intervals iced water intake, explain why there are changes in the temperature of the hypothalamus and the skin.**

On taking iced water, the temperature of the hypothalamus decreased and the temperature of the skin increased; because the iced water cools blood lowering the body core temperature; cooled blood flowing through the hypothalamus lowers the temperature of the hypothalamus; the thermo receptors in the hypothalamus detect the fall in body core temperature, in order to reduce heat loss from the skin by radiation, despite the warm chamber, due to the decreased body temperature; the hypothalamus is stimulated to transmit impulses from the thermo regulatory center to the vasomotor center in the medulla of the hind brain, stimulating it to fire impulses via the sympathetic nerves to the arterioles in the skin; constriction of the arterioles in the skin occurs forcing blood through the low resistant shunt vessels from arteries to veins, causing a lot of blood to by-pass the capillaries preventing heat loss; the decreased blood flow to the skin causes the temperature of the skin to rise as it approaches that of the hot environment of the hot chamber; the reduced rate of evaporation from the skin also causes retention of more heat further increasing the temperature of the skin.

**(ii) rate of energy loss by evaporation.**

On taking iced water, rate of energy loss by evaporation decreased gradually the iced water cools blood lowering the body core temperature, cooled blood flowing through the hypothalamus triggers thermo receptors in it which stimulates the heat gain center of the hypothalamus to send impulses through efferent nerves to different effectors to bring about changes to decrease heat loss to the environment so as to warm the body. The activity of sweat glands is reduced causing a reduction in rate of energy loss by evaporation.

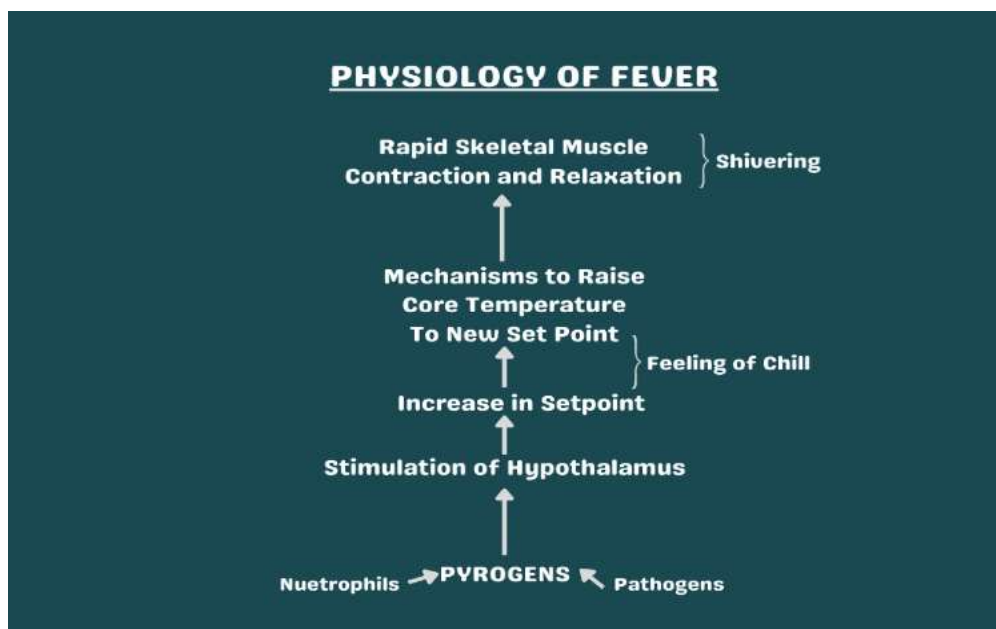
**(b) Why was iced water not given until 20 minutes after the start of the experiment?**

To allow the subject's body temperature equilibrate with that of his surrounding such that variations in body temperature as a result of swallowing ice can be clearly observed.

**(d) The onset of a fever is often accompanied by shivering and a feeling of cold known as chill.**

**(i) Explain these symptoms in terms of the mechanism of control of body temperature.**

Fever is caused when the set point of the thermostat that is the hypothalamus is raised. This is caused by substances known as pyrogens that may be produced by bacteria, or white blood cells known as neutrophils; With the set point raised, the normal body temperature is perceived as a low temperature as it is then lower than the new set point, causing a feeling of cold i.e. chill. The body continues to feel cold until the body temperature is equal to the new set point. The hypothalamus responds by causing mechanisms by sending impulses to effectors that cause mechanisms that bring about a rise in the core body temperature towards the new set point. In this case it stimulates the skeletal muscles to contract and relax, rapidly, a process accompanied by a high metabolic rate that involves generation of heat, the heat that raises the core body temperature to the new set point; the rapid contraction and relaxation is the shivering.



**(ii) What is the significance of this response to the body?**

The raised body temperature stimulates the defense responses of the body by speeding up the rate how cells work including the ones that fight illness and also make it harder for the virulent bacteria or virus to thrive.

**(e) (i) Outline the different ways in which plants avoid excess heat.**

- Folding of leaves
- Possession of a shiny waxy cuticle
- Wilting
- Transpiration
- Succulent leaves
- Plagiogeotropism

**(ii) Explain why plants suffer physiological damage when exposed to high temperature and high humidity.**

When plants are exposed to high humidity and high temperatures, the high humidity lowers the rate of transpiration; plants cannot lose heat by evaporation and therefore fail to reduce their temperature; The resultant increase in temperature above the optimum denatures enzymes which control the rate of metabolic reactions; this lowers the rate of metabolic reactions within plants.

**2(a) Compare cell-mediated response and humoral response.**

| <b>Cell mediated response</b>               | <b>Humoral response</b>                                       |
|---------------------------------------------|---------------------------------------------------------------|
| The whole cell attacks the pathogen         | Only antibodies are involved in the attack of pathogens       |
| Involves killing infected body cells.       | Body cells are not attacked                                   |
| T-cells cease being produced after weaning. | B-cells are continuously produced throughout one's life time. |
| Less specific                               | More specific.                                                |

**Similarities**

- Both involve secretion of chemical substance
- Both responses involve development of memory cells
- Both responses have their cells arising from the bone marrow.

**b) Describe the evidence that show that the thymus gland is important in the development of the immune response.**

- Removal of the gland from a newborn mouse result in death from a deficiency of lymphocytes in its tissue and blood.
- Tissue from an older mouse grafted onto an experimental new born mouse with the gland removed is unable to recognize and react with antigens.
- If the thymus gland is removed from a much older mouse, this mouse suffers no adverse effects.

**c) Explain how the immune response is regulated.**

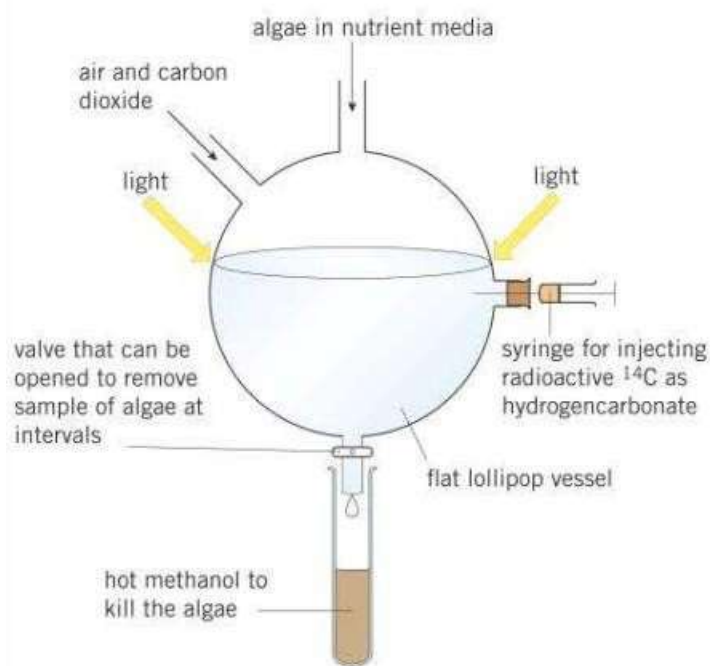
This is done by lymphokines secreted by T-suppressor and T-helper cells; in presence of foreign bodies, lymphokines produced by T-helper cells increases/boost immune response; for example, they cause B-cells to clone; these produce antibodies that eventually destroy the foreign material; the T-helper cell lymphokines also boost the activity of all other white blood cells such as phagocytes; lymphokines produced by T-suppressor cells decrease the immune response after the foreign bodies have been destroyed; they cut off the cloning of B-cells; so no antibodies are produced; they also reduce the activity of all other white blood cells; such as phagocytes.

**d) Explain why we do not produce antibodies against the antigens on our own.**

During development, the body carries out immune-orientation; during this process, the T-lymphocyte producing cells (stem cells of the bone marrow) must pass through the tissue of the thymus gland before becoming fully functional; as a result, any cells that recognize the body's own antigens are destroyed; such that later in life, the body does not produce antibodies against its own antigens.

---

3. **Melvin Calvin and his co-workers using a 'lollipop' experiment investigated the light-independent reaction. In this experiment single-celled algae are grown in the light in a thin-transparent 'lollipop'. Radioactive hydrogen-carbonate is injected into the 'lollipop'. This supplies radioactive carbon dioxide to the algae. At 5 seconds intervals, samples of photosynthesizing algae are dropped into hot methanol to stop chemical reactions instantly. The compounds in the algae are then separated out and those that are radioactive are identified. The 'lollipop' apparatus and a table summarizing the results of the experiment are shown below.**



▲ **Figure 7** The 'lollipop' apparatus used by Melvin Calvin

| Time (seconds) | Substances found to be radioactive                                        |
|----------------|---------------------------------------------------------------------------|
| 0              | carbon dioxide                                                            |
| 5              | glycerate-3-phosphate                                                     |
| 10             | glycerate-3-phosphate + triose phosphate                                  |
| 15             | glycerate-3-phosphate + triose phosphate + glucose                        |
| 20             | glycerate-3-phosphate + triose phosphate + glucose + ribulose biphosphate |

- (a) **Explain why the carbon dioxide supplied to the algae was radioactively labelled.**

To allow the substances to become incorporated to be identified.

- (b) **Explain how the information in the table provides evidence that glycerate-3-phosphate is converted into triose phosphate.**

The radioactive carbon is initially found in GP (5 seconds) and is next found in triose phosphate (10 seconds)

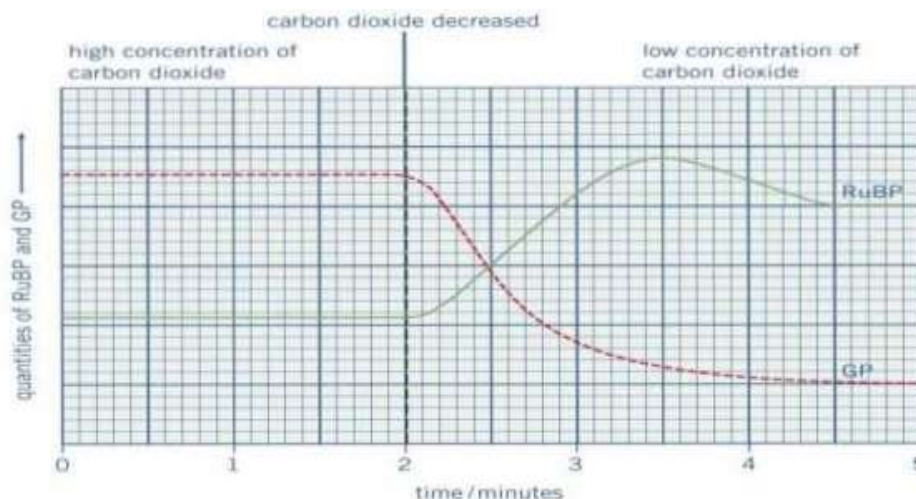
- (c) **Suggest an explanation of how methanol might stop further chemical reactions taking place.**

High temperature of the methanol denatures the enzymes that catalyze the reactions

(d) **Why was it necessary to conduct the ‘lollipop’ experiment in sufficient light conditions?**

Light stimulates the light dependent stage causing production of ATP and NADPH<sub>2</sub>. These compounds are essential in the conversion of GP to triose phosphate. ATP is also essential in the regeneration of RUBP.

**In a further experiment, samples of algae were collected at 1-minute intervals over a period of five minutes. The quantities of glycerate-3-phosphate (GP) and ribulose biphosphate (RuBP) were measured. At the beginning of the experiment, the concentration of carbon dioxide was reduced. The graph below shows the results of this experiment.**



(e) **Describe the effects on the quantities of GP and RuBP on the decrease of carbon dioxide after two minutes.**

Initially the concentration of G.P is high. But a decrease of CO<sub>2</sub> concentration causes a rapid decrease on the quantity of G.P. Initially the quantity of RUBP is low. But the decrease in CO<sub>2</sub> concentration causes a rapid increase in the concentration of RUBP.

(f) **Explain the changes in the levels of GP and RuBP.**

RUBP combines with CO<sub>2</sub> to form GP during the Calvin cycle/ light independent stage. GP is ultimately used to regenerate RUBP. When the CO<sub>2</sub> level is decreased there is less to combine with RUBP and hence less GP is formed but it is still being used up causing its levels to fall further. There is still some CO<sub>2</sub> and so some GP is made but much less than originally with less CO<sub>2</sub> to combine with RUBP it accumulates because it cannot be converted to GP. Its quantity rises to a new higher level due to lower levels of CO<sub>2</sub>.

(g) **Explain why the majority of plants are C<sub>3</sub> plants despite their disadvantage of photorespiration.**

C<sub>4</sub> photosynthesis has a higher energy cost than C<sub>3</sub> photosynthesis; they need relatively light intensities and temperature compared to C<sub>3</sub> plants; most regions have low light intensities and temperatures; e.g. temperate zones, high altitude area and some tropical area; these favour the growth of C<sub>3</sub> plants;

4(a) **Insect pests of crop plants can be controlled by chemical pesticides or biological agents. What are the advantages and disadvantages of using biological agents?**

**Advantages of using Bio control**

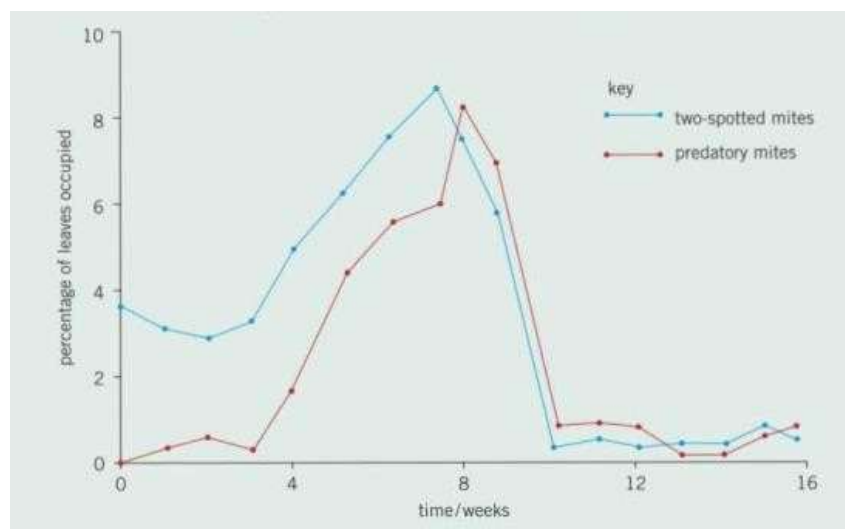
- It is an environmentally friendly method because it causes no pollution to non- target organisms.
- Self – perpetuating/ self-sustaining and hence permanent.

- Cost- effective.
- No artificial substances are added and hence animals that develop resistance against biological control agents are rare.

### Disadvantages of Bio control

- Potential agents are expensive to test for specificity.
- Host specificity testing may take several years to complete because of the need for thoroughness.
- Can sometimes fail in its specificity
- Expensive at the beginning
- May not completely destroy a pest

**Two-spotted mites are pests of strawberry plants. Ecologists investigated the use of predatory mites to control two-spotted mites. They then recorded the percentage of strawberry leaves occupied by two-spotted mites and by predatory mites over a 16-week period. The results are summarized in the figure below.**



**Explain the relationship between the percentage of leaves occupied by predatory mites and two-spotted mites during the period of this investigation.**

As the percentage of leaves occupied by two spotted mites increases the percentage of leaves occupied by predatory mites also increases this is because the predatory mites were still few in number therefore enabling the two spotted mites to increase in number and hence reproduce and multiply in number. Availability of enough food for the predatory mites enabling them to reproduce and multiply in numbers.

As the percentage of leaves occupied by the two spotted mites decreases; the percentage of leaves occupied by the predatory mites also decreases because many spotted mites were eaten by the predatory mites leading to decrease in their number which resulted into competition among the predatory mites for the few remaining spotted mites since they are source of food for the predatory mites, hence the predatory mites starved and some died leaving to their decreased number thus few occupied the leaves.



- (c) **The ecologists concluded that in this investigation the control of by the two-spotted mites by biological agent was effective. Explain how the results support this conclusion.**

Originally the percentage of leaves occupied by the two spotted mites was higher but after 10 weeks the percentage covered was reduced to very low levels

- (d) **Farmers who grow strawberry plants and read about this investigation might decide not to use these predatory mites. Suggest reasons for their decision**

The predatory mites did not completely eliminate the two spotted mites.

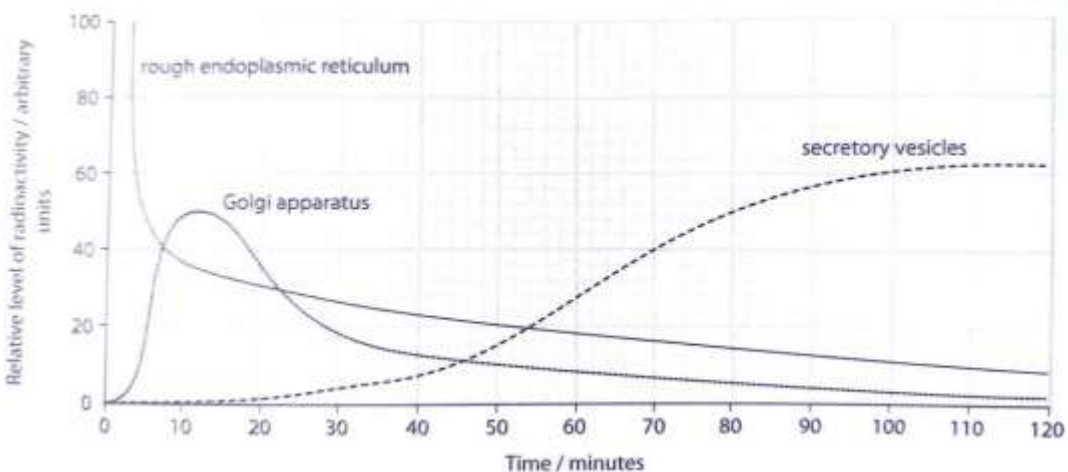
On third week there was a rapid increase in the percentage of leaves covered with the two spotted mites.

The predatory mites were slow at adapting to the new environment

- (e) **The ecologists repeated the investigation but sprayed chemical pesticide on the strawberry plants after 10 weeks. After 16 weeks no predatory mites were found but the population of two-spotted mites had risen significantly. Give an explanation for the rise in the population of two-spotted mites.**

The insecticide successfully killed the pests (the two spotted mites) but also targeted the biological control agent (the predatory mite) at the same time. The rise in the two- spotted mites is attributed to the elimination of biological control agent by pesticides or moving to other places to seek for new prey

5. **An investigation was carried out to track the movement of amino acids through different organelles in a cell. Pancreatic cells were provided with a small quantity of amino acids containing tritium atoms,  $^3\text{H}$ , instead of normal hydrogen atoms. Tritium is radioactive, and molecules containing it can be detected. They are said to be tritium 'labelled'. After a short-time the pancreatic cells were provided with a large quantity of unlabeled amino acids. Samples of cells were extracted at particular times, and the levels of radioactivity in the rough endoplasmic reticulum, Golgi apparatus and secretory vesicles were determined. The results are summarized in the figure below.**



**(a) Explain the pattern shown by these results.**

From 10 minutes to 120 minutes the relative level of radioactivity in the secretory vesicles increase gradually because the proteins in the Golgi vesicles are being transported to the secretory vesicles to be secreted out of the pancreatic cell.

**(b) Explain why the pancreatic cells were chosen for this experiment.**

Pancreatic cells secrete variety of substances made of proteins e.g. hormones, enzymes at a rate high enough that they have extensive organelles like the Golgi apparatus, rough endoplasmic reticulum and many secretory vesicles which makes the tracing of the radioactive amino acid easy

**(c) With evidence from the figure above state the sequence in which labelled amino acids moved through the three organelles.**

The amino acids moved through the rough endoplasmic reticulum, Golgi apparatus then to the secretory vesicles because as the level of radioactivity decreases in the rough endoplasmic reticulum, the relative level of radioactivity increases in the Golgi apparatus decreases the relative level of radioactivity in the secretory vesicles increases gradually

**(d) Insulin does not enter liver cells but brings about its effects with a receptor on the liver cell surface.**

**(i) Explain why insulin molecules are not able to enter liver cells.**

Insulin is polar and the cell surface membrane has a lipid bilayer which is non polar therefore insulin cannot pass through the phospholipids of the liver cell surface membrane of the liver cell so insulin will only work by binding to the receptor sites in the liver cell without entering the liver cell in order to bring about its effects

**(ii) What kind of molecules in a cell membrane that could act as a receptor for insulin?**

Glycoproteins because proteins have very specific shapes with receptor sites to which insulin bind.

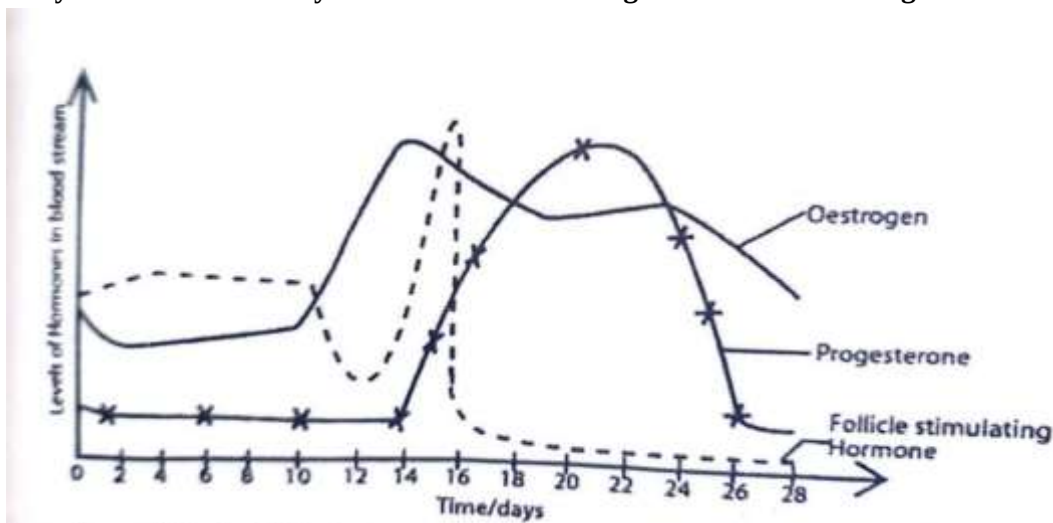
**(iii) Explain how the binding of insulin on the surface membrane of liver cells causes a decrease in blood glucose concentration.**

Insulin binds onto the receptor sites of the liver cells increasing permeability of the cell surface membrane to glucose. More glucose is taken in the liver cells.

Increase rate of conversion of glucose to glycogen in the liver cell to be stored.

Insulin also decreases the rate of gluconeogenesis

6. A study was conducted on the hormonal changes in the menstrual cycle of a fertile female of a reproductive age. Periodical changes in the level of the gonadotrophic hormone, follicle stimulating hormone and two other steroid hormones, oestrogen and progesterone were studied over 28 days of the menstrual cycle. The hormonal changes are shown in the figure below.



- (a) Compare the variation in the levels of oestrogen and progesterone.

#### Similarities

- In both, the hormonal levels decreased from 0 to 2 days and from 23 days to 26 days.
- In both, the levels of the hormones increased to reach a peak;
- In both the levels of hormones reached peak maximum.
- In both levels of the hormones declined after the peak;
- The levels of the hormones are the same on 17<sup>th</sup> day and 23<sup>rd</sup> day.

#### Differences

| • Levels of oestrogen hormone                                                      | • Level of progesterone hormone                         |
|------------------------------------------------------------------------------------|---------------------------------------------------------|
| • Increase very slowly from 2 to 10 days and then increases rapidly up to 13 days; | • Remained constant from 2 up to 13 days                |
| • Decreased rapidly from day 14 up to 19 days                                      | • Increased rapidly from 14 up to about 19 days         |
| • Two peaks reached                                                                | • One peak reached                                      |
| • Declined rapidly from day 26 up to 28 days                                       | • Remained constant from day 26 to day 28               |
| • Levels higher from 0 to 17 days and from 24 to 28 days                           | • Levels lower from 0 to 17 days and from 24 to 28 days |
| • Levels lower from 17 up to 24 days                                               | • Levels higher from 17 up to 24 days                   |
| • Attains peak earlier on day 13                                                   | • Attains peak later on day 21                          |

(b) Explain the

- (i) relationship between follicle stimulating hormone and oestrogen in the 28 days of the menstrual cycle.
- (ii) variations in the level of progesterone in throughout the cycle.
- (c) (i) Predict changes in the level of the three hormones if fertilization occurs on the 17<sup>th</sup> day of the cycle.
- (ii) Suggest reasons for your answer in (c)(i) above.

**(i) Relationship between follicle stimulating hormone and Oestrogen in the 28 days of the menstruation cycle**

From (1 to about 3 days: levels of the follicle stimulating hormone (FSH) increased gradually while the levels of Oestrogen decreased gradually; decrease in levels of Oestrogen stimulates hypothalamus to produce gonadotroph releasing hormone (GnRH). Gonadotropin releasing hormone stimulates secretion of FSH; from the anterior lobe of pituitary gland. High levels of FSH inhibits secretion of the oestrogen from ovarian follicles by negative feedback from 3 up to 10 days; Oestrogen increases very slowly while FSH remain almost constant; but the levels of FSH remain relatively higher than Oestrogen; because FSH stimulates development of few ovarian follicles which secrete little amounts of Oestrogen and from 10 up to 13 days, levels of FSH decreased rapidly while the levels of Oestrogen increased rapidly because FSH stimulated development of very many ovarian follicles which secrete large amounts of Oestrogen. Oestrogen causes repair and growth of endometrium; and inhibits further secretion of follicle stimulating hormones (FSH) so that only one graafian follicle develops within the ovary. From 13 up to 15 days the levels of FSH increased very rapidly to a peak while levels of Oestrogen declined slowly because levels of Oestrogen had reached its peak and it stimulated hypothalamus to secrete more gonadotropin releasing hormone causing a temporary rapid increase in secretion of FSH and luteinizing hormone (LH) from the anterior lobe of pituitary gland.

Decline gradually up to day 20 where levels of FSH remained constant; high levels of Oestrogen and increase in concentration of progesterone; inhibit secretion of FSH from anterior lobe of pituitary gland; FSH levels at its peak inhibit secretion of Oestrogen from the ovaries. From 19 up to 24 days; levels of Oestrogen increased gradually while levels of FSH remain very low and constant: this is because corpus luteum secretes smaller amounts of progesterone. From 24 up to 28 days levels of progesterone inhibiting secretion of FSH remained constant corpus luteum degenerates secretion of Oestrogen from it cause and there continues to be no further secretion of FSH from anterior lobe of pituitary gland caused by the inhibitory effects of progesterone and Oestrogen:

**(ii). Variation in the levels of progesterone hormone throughout the menstrual cycle**

At 0 days, the levels of progesterone is very low and remained almost constant up to 2 days from 2 up to 14 days; levels of progesterone remained constant within low levels; corpus luteum is not yet developed and no further secretion of progesterone into the blood stream. From 14 days to about 17 days, levels of progesterone increased rapidly because the corpus luteum formed is secreting large amounts of progesterone hormones; From 17 up to 21 days, the levels of progesterone increases gradually because the size of the corpus luteum begins to reduce and the levels of progesterone secreted reduces. From 21 up to 26 days, levels of progesterone declined rapidly because corpus luteum is degenerating rapidly and further secretion of progesterone is inhibited/ reduced greatly, Complete degeneration of the corpus luteum causes the levels of progesterone to remain constant up to 28 days.

**(c) Predict the changes in the level of the three hormones if fertilization occurs on the 17<sup>th</sup> day of the cycle**

- Level of progesterone will increase gradually and then rapidly above the levels of Oestrogen
- Levels of Oestrogen will increase very gradually but slightly below the levels of progesterone
- Levels of follicle stimulating hormone will decreased gradually and remains constant at very low levels

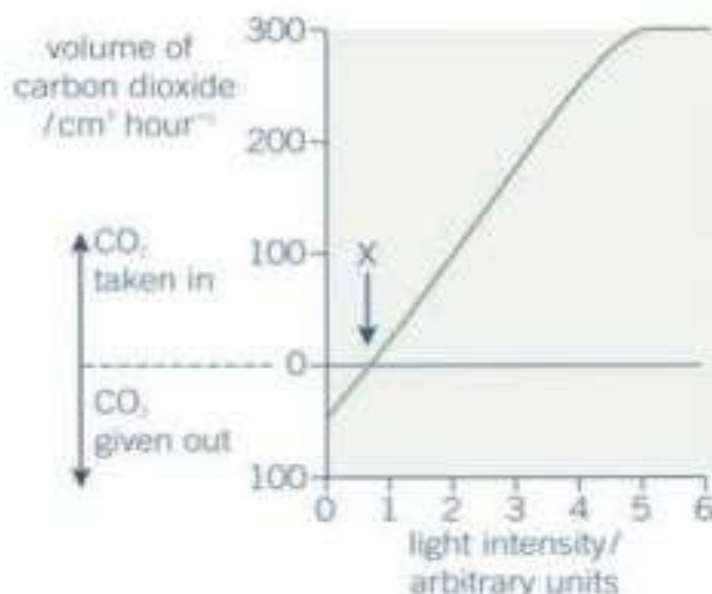
**(c) (i) Suggest the reasons for your answer in (c) above**

Diploid zygote formed will develop into a blastocyst, the cells of the outer layer of the blastocyst called trophoblasts secrete Human chorionic gonadotropin hormone which will maintain the corpus luteum for some time secreting small amounts of Oestrogen but larger amounts of progesterone a role later played by the placenta when formed. Progesterone maintains the thickness of endometrium which sustains pregnancy; oestrogen stimulates formation of receptor molecules in the myometrium; sensitive to oxytocin; that will later cause birth; FSH decreases to ensure no ovulation occurs during pregnancy/No development of graafian follicles take place during pregnancy.

**(d) Explain the significance of the three types of hormones as fertility drugs**

They can be provided to mature female individuals as synthetic chemicals that can stimulate ovulation they either provide gonadotropins such as FSH which stimulates the development of follicles or they provide a chemical which inhibit natural production of Oestrogen. Lack of Oestrogen results in production of more FSH which stimulates follicular development.

7. **The figure below shows the volume of carbon dioxide produced by a sample of tomato plants at different light intensities.**



**(a) Explain the effect of light intensity on carbon dioxide exchange in the tomato plants.**

Increase in light intensity from 0 to 0.8 arbitrary units causes a rapid decrease in carbon dioxide released; because light causes process of photosynthesis to take place, using up much of CO<sub>2</sub> released by the plant during respiration; increase in light intensity from 0.8 to 5 arbitrary units causes a rapid increase in carbon

dioxide absorbed; because, more stomata open; and with wider aperture; taking up more carbon dioxide; also, more ATP and NADPH<sub>2</sub> are produced during light dependent reactions; thus more carbon dioxide is fixed in the Calvin cycle; causing more to be absorbed along a concentration gradient; increase in light intensity beyond 5 arbitrary units, causes no effect in the carbon dioxide taken in; because, light is no longer a limiting factor in process of photosynthesis; other factors such as temperature are now limiting.

**(b) Explain why at point X neither carbon dioxide is not taken in nor given out.**

- The rate of respiration is equal to the rate of photosynthesis;

**c) Some herbicides cause the stomata of plants to close. Explain how these herbicides might lead to the death of plants.**

With stomata closed, there is little gaseous exchange with the environment; while there will still be some interchange of gases produced by respiration and photosynthesis; neither process can continue indefinitely by relying exclusively on the gases must be obtained from the environment; In the absence of this supply both photosynthesis and respiration will cease and the plant will die

**(d) Explain the advantage to a plant being able to control the opening and closing of the stomata.**

Enables the plant to control the amount of water lost through stomatal transpiration; for example; closing stomata during water stress so that the plant does not dry out as a result of water loss; opening the stomata in times of reduced temperature so that the plant can take up carbon dioxide for photosynthesis; and lose some water in transpiration to cool the plant; and take up more water using the transpiration stream

**(e) Water flow over the fish gills is one-way whereas the flow of air in and out of the mammalian lungs is two-way. Explain why the one-way flow of a fish is advantageous over the two-way flow in the mammalian lungs.**

In fish, a steep diffusion gradient is maintained over the whole length of the gill lamellae; in mammalian lungs, fresh air inhaled mixes with stale air in the dead space; reducing the diffusion gradient;

**(f) Outline the physiological effects in the body of breathing in excess**

**(i) carbon dioxide**

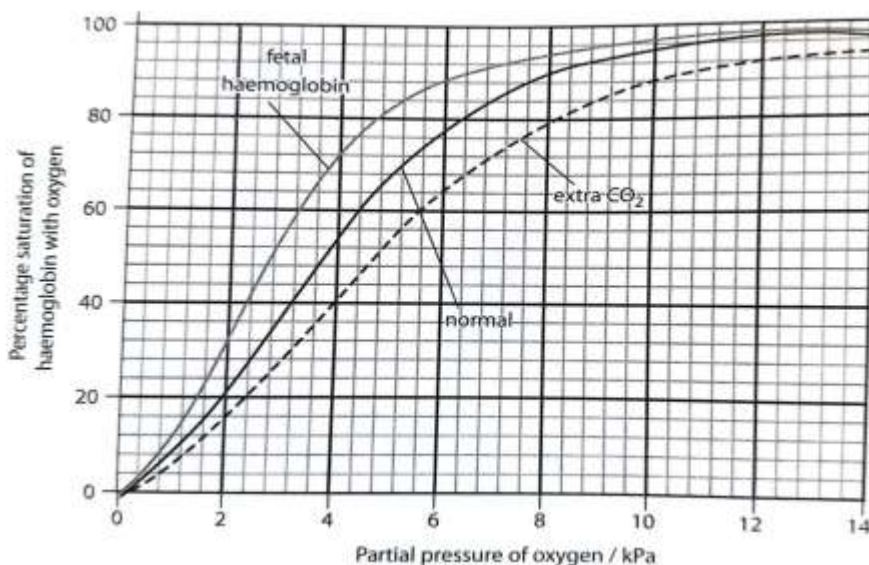
In excess carbon dioxide; there is low pH which encourages the dissociation of oxyhemoglobin to release oxygen that will be utilized by the body tissues;

Excess carbon dioxide also interferes with tissue respiration; whereby it increases the acidity of blood and tissue fluids; inhibiting enzymes and stopping essential metabolic process

**(ii) oxygen.**

Breathing in excess oxygen at atmospheric pressure presents no problems; this because at this pressure, breathing in excess oxygen does not increase the amount of oxygen delivered to the tissues; but when excess oxygen is breathed in at a pressure greater than atmospheric pressure, the tissues metabolize rapidly to keep pace with the excess oxygen supply; build up of excess oxygen in the body inhibits some enzymes involved in the Krebs' cycle, interfering with cell respiration/metabolism; this may result into muscle twitching followed by nausea, dizziness, impaired hearing, impaired vision, breathing difficulties, confusion, lack of coordination, convulsions and death.

8. The figure below shows the oxygen dissociation curves of human haemoglobin in the presence of a range of concentrations of carbon dioxide.



- (a) Explain how the structure of the haemoglobin molecule generates an oxygen dissociation curve that is sigmoid.

At very low concentration of oxygen, the four polypeptides of haemoglobin molecule are closely packed making it difficult to absorb the first molecule of oxygen, hence the slow increase in percentage saturation of haemoglobin with oxygen.; after loading with the first oxygen molecule the haemoglobin molecule changes shape, making it easier for the remaining three oxygen molecules to be loaded; hence the rapid rise in percentage of saturation of haemoglobin with oxygen; at higher partial pressure of oxygen with haemoglobin becomes fully saturated with oxygen, hence the levelling of the curve.

- (b) (i) The partial pressure of oxygen in the alveoli 12 kPa. Use the graph to determine the percentage saturation of normal haemoglobin in the alveoli.

98%

- (ii) 1 g of fully saturated haemoglobin carries 1.3cm<sup>3</sup> of oxygen. Calculate the volume of oxygen bound to 1 g of haemoglobin at a partial pressure of 12 kPa.

1.27

- (ii) Use the graph to determine the volume of oxygen released to rapidly respiring cells.

1.036cm<sup>3</sup>

- (b) Why is it an advantage to the body for the dissociation curve to take the orientation shown on the graph at a high carbon dioxide concentration?

Respiring muscle will have higher carbon dioxide concentration; this reduces the affinity of haemoglobin for oxygen; so more oxygen is released to the muscle.

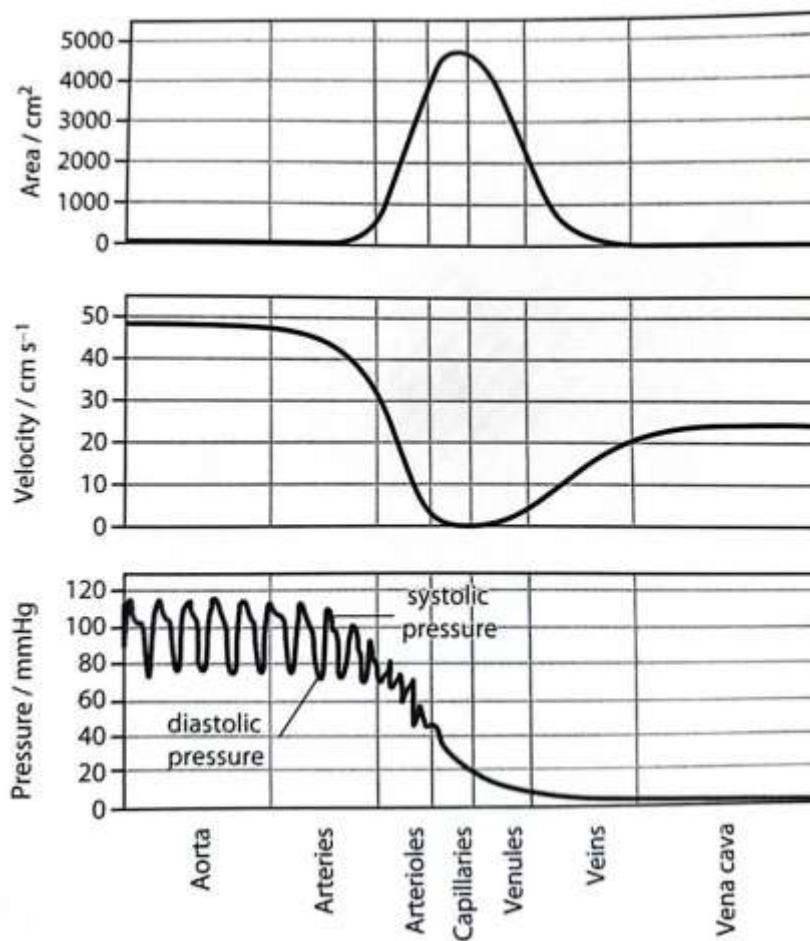
- (d) Explain why the foetal oxygen dissociation curve is different to that of the normal adult haemoglobin.

Foetal haemoglobin has higher affinity for oxygen; this ensures that oxygen will diffuse from the maternal to the foetal blood

(e) Explain how the of red blood cells maximizes the transport of oxygen to tissues.

- They have a small size and flexible so that they can fit through narrow blood capillaries
- They have a biconcave shape which maximizes their surface area to absorb oxygen
- Have thin membrane so that gases can diffuse through
- Contain haemoglobin which binds to oxygen
- They have no nucleus so they can contain more haemoglobin

9. The figures below show how blood pressure, cross-sectional area and blood velocity change in different blood vessels.



(a) (i) Explain how the blood pressures change as blood moves from the aorta to the vena cava.

High pressure pulsatile in the aorta, arteries, arterioles, due to close proximity to the heart; pressure rapidly drops as blood flows through capillaries; due to increased capillary cross sectional area as a result of extensive branching; pressure lower in venules; veins and vena cava and not pulsatile; this is because, they are far away from the heart/ventricles.



**(ii) Explain how the velocity changes as blood flows from the aorta to the vena cava.**

Velocity is high in the aorta due to proximity to the heart, velocity falls or decreases rapidly through arteries and arterioles; due to increased distance from the heart; also, due to reduced pressure as the total volume of the blood vessel decreases due to dividing to form arteries, arterioles and capillaries; also, velocity drops due to resistance to the blood flow (friction) of the narrowing lumen of blood vessels; velocity rises gradually in venules, veins and vena cava; due to wider lumen of veins; which offer less resistance to blood flow; also due to skeletal muscle contraction; thoracic pressure due to inspiration.

**b) When an animal is wounded, its overall blood pressure rises, but the area in the vicinity of the wound swells as a result of local vasodilation. Suggest what the advantage of these changes might be.**

Local vasodilation in the wounded area enables more blood carrying oxygen and nutrients to arrive there and speed up the process of repair and replacement of tissues; increased blood pressure prepares the body of the animal to respond to any further stress readily and efficiently;

**(c) Describe the role of the nervous and hormonal systems in the control of the heart rate.**

- The Centre responsible for nervous control of the heart rate is located in the medulla oblongata; the medulla contains the cardiac inhibitory Centre which reduces the heart rate and the cardiac accelerator Centre which stimulates the heart rate.
- Vagus (parasympathetic nerve) from the cardiac inhibitory centre innervate the SAN AVN and the bundle of his; impulses passing through the vagus nerves reduce the heart rate.
- Sympathetic nerves from cardiac accelerator Centre of the medulla innervate the SAN and increase the heart rate;
- Sensory nerve fibres from stretch receptors within the walls aortic arch, carotid sinuses and the vena cava run to the cardiac inhibitory Centre in the medulla; impulses from the aorta and carotids decrease the heart rate while those from the vena cava stimulate the accelerator Centre which increases the heart rate.
- The hormonal control involves adrenaline (and small amounts of noradrenaline) from the adrenal medulla; both stimulate the heart rate; thyroxine from the thyroid gland stimulates the heart rate.

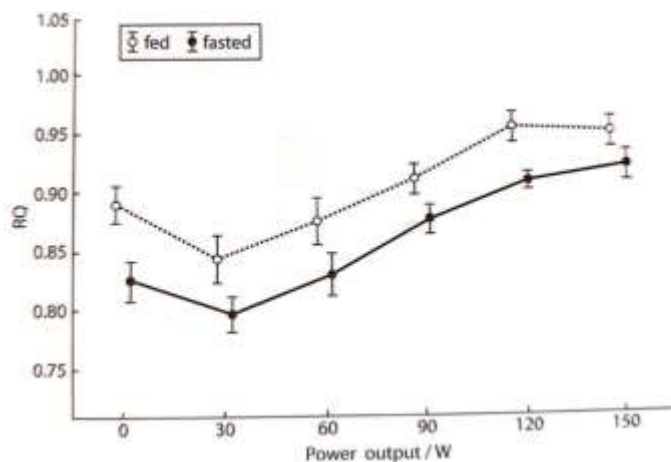
**(d) Outline the adjustments that occur to the heart rate and circulatory system just before, during and after a vigorous exercise.**

**Before the vigorous exercise;** Adrenaline hormone is secreted in anticipation of the race; this stimulates vasoconstriction throughout the body in all but the most vital organs; this results in increased blood pressure; heart rate is also increased; extra blood is also passed to the general circulation from the spleen;

**During the race;** increased metabolic activity takes place, especially the skeletal muscles; increased carbon dioxide levels in these regions promote local vasodilation; increased body temperature further enhances vasodilation; the general increase in carbon dioxide level in blood is detected by chemoreceptors of the aorta and carotid bodies which in turn stimulate the vasomotor centre to promote vasoconstriction; blood pressure is thus increased and blood flows faster; heart rate is also increased and a more emptying of the heart ventricles occurs; towards the end of the exercise, the muscles respire anaerobically and so lactic acid is produced; strong contraction of the muscles squeeze the veins and promote faster return of blood to the heart via veins.

**After the exercise;** the oxygen debt is paid off and lactic acid removed from the blood system; tissues subside in activity and the carbon dioxide level decreases; consequently, there is a return to normal heart rate, and blood pressure.

- 10 An investigation was carried out into the effect of fasting on the RQ of two students. One of the students was given no food (fasting) for 12 hours while the other ate a normal balanced diet. The students were then made to exercise at increasing intensities in rooms set at the same temperature and humidity. The volumes of oxygen used in their bodies and carbon dioxide produced were measured and the RQ calculated for the different exercise intensities. Intensity was measured as power output. The results are shown in the figure below.



- (a) (i) The temperature and humidity of the room were kept the same. Suggest two factors that would need to be controlled.

Oxygen partial pressure  
Carbon dioxide partial pressure

- (iii) Compare the effects of increasing exercise intensity on RQ of the two students.

#### Differences

The RQ for the students that were fed is higher than that of students who fasted; beyond 120 power output, the RQ for the fed students was constant whereas that for the students who fasted increased gradually

#### Similarities

In both student groups, the RQ decreased during the first 30 power output; in both student groups, the RQ increased between 30 and 120 power output

- (iv) Explain the results obtained in the figure above.

Fed students used more aerobic respiration of carbohydrates; fasted students have a lower glycogen reserve; so they utilized more fat for energy production; both groups used a mixture of carbohydrate and fat; both begin utilizing aerobic respiration of carbohydrates; the increase of RQ may be due to increasing use of carbohydrate or proteins or anaerobic respiration; as intensity of the exercise increases, anaerobic respiration increases/ or aerobic respiration decrease.

- (b) Migratory birds like geese have a much higher ratio of saturated fatty acid than non-migratory birds such as ostriches. Explain the reasons for this difference.

This is because fatty acids are lighter; so they do not impose much weight on the flying bird during migration; also, fatty acids produce more energy per gram when oxidized; as they contain more hydrogen; this great deal of energy is needed for flying during migration over the long distances.

**c) Most hibernating animals have an RQ between 0.8 to 0.9. Give an explanation for this observation.**

During hibernation, the animal is resting but not starved and protein is not a substrate for respiration during these conditions; the observed values are due to a mixture of carbohydrate (1.0) and lipid (0.7) being respired

**(i) Describe the process of ATP synthesis in the mitochondrion.**

Hydrogen ions released from glycolysis and the Krebs cycle are picked by hydrogen carrier in the mitochondrial matrix; hydrogen atoms then split into protons( $H^+$ ) and electrons ( $e^-$ ); the electrons go through the electron transfer chain; The protons driven by energy from the electron transfer chain are pumped into the inter membrane space creating a lower pH in the inter-membrane space than in the matrix.

This creates asleep electrochemical gradient across the inner mitochondrial membrane; Protons diffuse back into the matrix down their gradient via the stalked particles; As they diffuse back energy from them is used to synthesize ATP from ADP and inorganic phosphate. The reaction is catalyzed by ATPase enzyme in the stalked particle. (Ref; Michael Roberts, Michael Reis and Grace Monger-Advanced Biology pg 145)

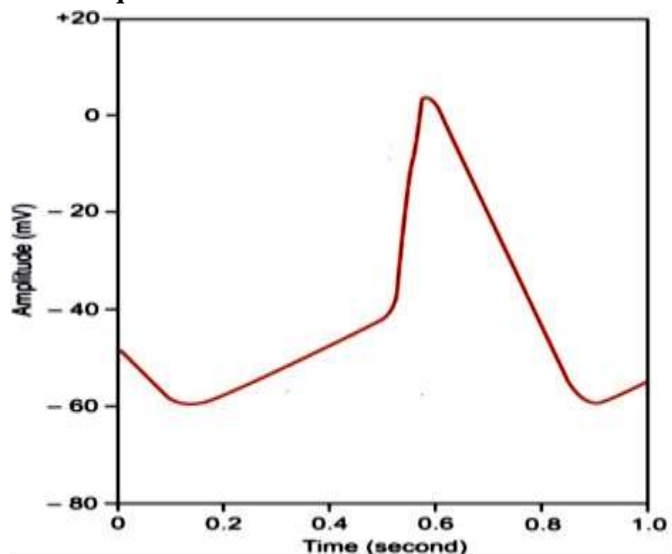
**(ii) Outline the ways in which ATP is used by the cells.**

- Synthesis of complex substances from simpler ones; e.g., polypeptides from amino acid, nucleic acid, nucleotides, etc
- Active transport of substances against a diffusion gradient e.g., activity of sodium potassium pump
- Mechanical work such as muscle contraction 'movement by means of cilia and flagella, movements of vesicles through the cytoplasm
- Bioluminescence (emitting of light by living organisms) and electrical discharge

**(e) Explain why the majority of plants cannot survive in flooded soils but rice can.**

- Some varieties of rice respond to flooding by growing taller; this allows exchange of gases through the stomata of leaves to occur;
- Rice stems contain loosely packed aerenchyma cells; gases are able to diffuse through the aerenchyma cells
- Their cells in the roots are tolerant to alcohol formed as result of the anaerobic respiration (alcoholic fermentation); they produce an enzyme **alcohol dehydrogenase** which breaks down ethanol; Thus allows the plant to grow actively in conditions of oxygen shortage using ATP produced from alcoholic fermentation.

- 11(a) The figure below shows the variation of the amplitude (potential) of the SAN of human cardiac muscle (pacemaker) with time. The resting potential of the SAN is -55 mV. Study the graph carefully and answer questions that follow.



- (i) Describe the changes in the potential/amplitude of the SAN with time.

From 0s to 0.15s amplitude decreased rapidly to minimum; from 0.15s to 0.55s amplitude increased gradually; from 0.55s to 0.6s amplitude increased very rapidly to a peak/maximum; from 0.6s to 0.9s amplitude decreased rapidly to a minimum; from 0.9s to 1.0s amplitude increases gradually;

- (ii) Account for the above changes in the figure above

| Range                                                                 | Reason                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| From 0s to 0.15s amplitude decreased rapidly to minimum               | Unstable/not stable resting potential of the SAN because potassium ion channels are open; potassium ions diffuse out creating a more negative potential;                                                                                                                                                                                       |
| From 0.15s to 0.55s amplitude increased gradually                     | Because calcium and sodium, voltage gated channels open slowly; resulting into a gradual influx of sodium and calcium ions; causing gradual depolarization of pacemaker cells; to the threshold level at -40mV;                                                                                                                                |
| From 0.55s to 0.6s amplitude increased very rapidly to a peak/maximum | After reaching threshold level; triggers rapid opening of more calcium and sodium ions; leading to rapid influx of sodium and calcium causing rapid depolarization;                                                                                                                                                                            |
| From 0.6s to 0.9s amplitude decreased rapidly to a minimum;           | Due to closure of calcium and sodium ion channels and opening of potassium voltage channels; efflux of potassium ions; but potassium ion channels remain open for a long period of time resulting into more efflux of potassium ion causing the membrane potential to fall/decrease below resting potential; hyperpolarization for short time; |

|                                                         |                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>From 0.9s to 1.0s amplitude increases gradually;</i> | <i>Slow influx of sodium/calcium ions leading to gradual depolarization/development of pacemaker potential which trigger the next action potential (Electrical potential in SA node is different from that of other cardiac muscle fibers. In SA node, each impulse triggers the next impulse)</i> |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**(b )(i) Explain the effects of the different innervations to the SAN.**

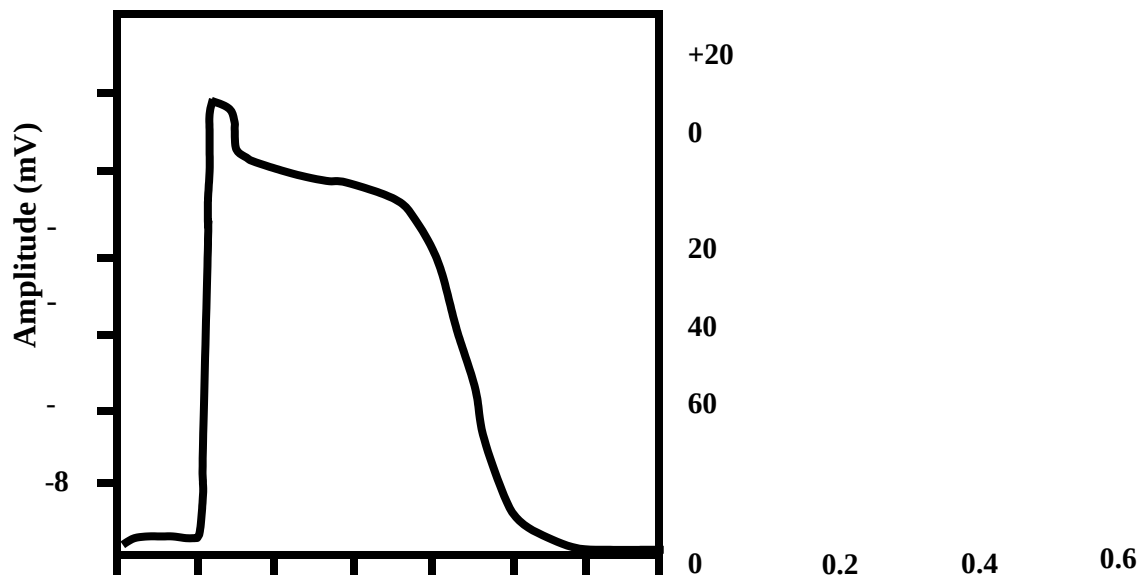
Sympathetic nerves; release noradrenaline/Norepinephrine into pacemaker cells; increasing permeability of the  $\text{Na}^+$  and  $\text{Ca}^+$  ion channels; rapid influx of calcium and sodium ions; results into resting potential of the pacemaker drift up more rapidly; interval between action potential decreased and heart beats up faster;

Parasympathetic nerves; release acetylcholine (ACh); into pacemaker cells; increasing permeability of  $\text{K}^+$  channels; efflux of potassium ions; membrane potential becomes more negative following an action potential; and rises more slowly; ACh also decreases permeability of  $\text{Ca}^+$  and  $\text{Na}^+$  so that the rate of rise of membrane potential slow; interval between pacemaker potentials lengthens; and slows heart rate;

**(ii) Explain the functions of the conducting systems of the heart**

| <i>Range</i>                                                                              | <i>Reason(s)</i>                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>From 0s to 0.05s, amplitude increased rapidly to a peak</i>                            | <i>Rapid opening of voltage gated <math>\text{Na}^+</math> ion channels; rapid influx of sodium ions; leading to rapid rate of depolarization;</i>                                                                                                                                                                                                                                                             |
| <i>from 0.05s to 0.1s, amplitude decreased rapidly;</i>                                   | <i>Rapid closure of sodium voltage gated channels and opening of potassium ion channels causing short repolarization;</i>                                                                                                                                                                                                                                                                                      |
| <i>from 0.1s to 0.3s amplitude remained almost constant; from 0.3s to 0.4s, amplitude</i> | <i>Slow opening of calcium/sodium ion channels; which are kept open for a long period of time; to sustain entry of large number of calcium ion required for contractile process of ventricles;</i>                                                                                                                                                                                                             |
| <i>from 0.3s to 0.4s, amplitude decreased rapidly to resting potential;</i>               | <i>Rapid opening of potassium ion channels resulting into rapid efflux of potassium ions; number of potassium ions moving out of the muscle fiber exceeds the number of calcium ions moving in</i>                                                                                                                                                                                                             |
| <i>from 0.4s to 0.6s amplitude remained almost constant;</i>                              | <i>At the end of final repolarization, all sodium ions, which had entered the cell throughout the process of action potential; move out of the cell and potassium ions move into the cell; by activation of sodium-potassium pump; Simultaneously, excess of calcium ions, which had entered the muscle fiber also move out through sodium-calcium pump. Thus, the resting membrane potential is restored.</i> |
|                                                                                           | <i>”</i>                                                                                                                                                                                                                                                                                                                                                                                                       |

- (c) The graph below shows changes in the ventricular action potential. Study the graph carefully and answers that follow.



- (i) From the graph above describe changes in the ventricular action potential.

From 0s to 0.05s, amplitude increased rapidly to a peak; from 0.05s to 0.1s, amplitude decreased rapidly; from 0.1s to 0.3s amplitude remained almost constant; from 0.3s to 0.4s, amplitude decreased rapidly to resting potential; from 0.4s to 0.6s amplitude remained almost constant;

(ii) Explain the changes in the ventricular action potential.

| <b>Range</b>                                                                                     | <b>Reason(s)</b>                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>From 0s to 0.05s, amplitude increased rapidly to a peak</i></b>                            | <i>Rapid opening of voltage gated Na<sup>+</sup> ion channels; rapid influx of sodium ions; leading to rapid rate of depolarization;</i>                                                                                                                                                                                                                                                                       |
| <b><i>from 0.05s to 0.1s, amplitude decreased rapidly;</i></b>                                   | <i>Rapid closure of sodium voltage gated channels and opening of potassium ion channels causing short repolarization;</i>                                                                                                                                                                                                                                                                                      |
| <b><i>from 0.1s to 0.3s amplitude remained almost constant; from 0.3s to 0.4s, amplitude</i></b> | <i>Slow opening of calcium/sodium ion channels; which are kept open for a long period of time; to sustain entry of large number of calcium ion required for contractile process of ventricles;</i>                                                                                                                                                                                                             |
| <b><i>from 0.3s to 0.4s, amplitude decreased rapidly to resting potential;</i></b>               | <i>Rapid opening of potassium ion channels resulting into rapid efflux of potassium ions; number of potassium ions moving out of the muscle fiber exceeds the number of calcium ions moving in</i>                                                                                                                                                                                                             |
| <b><i>from 0.4s to 0.6s amplitude remained almost constant;</i></b>                              | <i>At the end of final repolarization, all sodium ions, which had entered the cell throughout the process of action potential; move out of the cell and potassium ions move into the cell; by activation of sodium-potassium pump; Simultaneously, excess of calcium ions, which had entered the muscle fiber also move out through sodium-calcium pump. Thus, the resting membrane potential is restored.</i> |
|                                                                                                  | ”                                                                                                                                                                                                                                                                                                                                                                                                              |

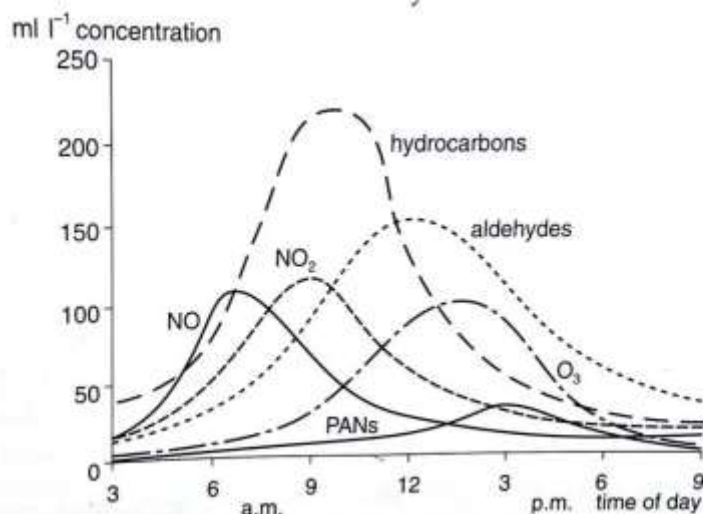
(iii) Account for the significance of the changes represented above

Initial rapid depolarization and sustained depolarization is due to entry of sodium ions into the cardiac muscle; depolarization leading to propagation of action potential; release of calcium ions from sarcoplasm reticulum; calcium ions bind with troponin; reversing the inhibitory effect of tropomyosin; myosin binding site on actin is exposed; bind forming actomyosin cross-bridge; release of energy accompanies cross-bridge formation; myosin filament pulls actin filament over itself resulting into contraction of heart muscle during systole to eject blood out of the ventricles;

Repolarization reduces muscle's potential, due to closure of sodium gates, but potassium ion gates open, activates calcium pump to actively move Ca<sup>+</sup> ions from sarcoplasm into sarcoplasm reticulum; where they are stored to wait for the next action potential to trigger their release; all this terminates the ventricular systole to allow the ventricles with blood during ventricular diastole;

Restoration of resting potential to prepare the muscle cells ready for the next action potential; for proper functioning;

12(a) The graph below shows the composition of the photochemical smog on a sunny day in Kampala city. Study it carefully and answer questions that follow.



(i) **Explain why the concentrations of NO, NO<sub>2</sub> and Hydrocarbons increase in the morning.**

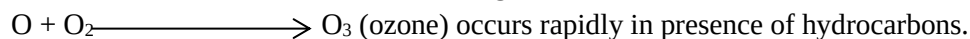
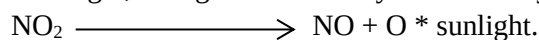
In the morning there are increased number vehicles entering Kampala; because people are heading to work; emissions of pollutants from petrol engines; in the morning weather is calm and cold resulting into cold air which traps pollutants; cold air is denser; so rise less easily; so the pollutants do not disperse easily;

(ii) **Suggest an explanation for the timing of the peak concentrations of NO and NO<sub>2</sub>.**

The concentration of NO peaks earlier at 6am while NO<sub>2</sub> peaks later at 9am. This is because Nitrogen monoxide is oxidized to nitrogen dioxide with gradual increase in temperature /sunlight; oxidation catalysed by the presence of Hydrocarbons and CO; peak in early morning because this is the time many cars/automobiles/vehicles enter the city; and their engines discharge these oxides

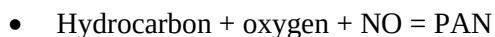
(iii) **Explain how hydro-carbons accelerate the formation of Ozone hole.**

In presence of sunlight, nitrogen dioxide may react with hydrocarbons to form tropospheric ozone;



(iv) **Explain the increase in the concentration of PANs (peroxyacetyl nitrate) in the mid-morning.**

PAN (peroxyacetyl nitrates) formed when aldehydes and ketones react in presence of sunlight to form radicals; radicals react with NO and NO<sub>2</sub> to form PAN;



(v) **Why might cyclists advised not to leave work early in order to avoid rush-hour traffic.**

Evening rush hour traffic from 6pm to 7pm, results into a rapid buildup of NO and NO<sub>2</sub>; due to emission from automobiles; and their reaction to form secondary pollutants such as PAN; these pollutants are inhaled causing respiratory problems/Asthma/throat/eye irritants and all these are life threatening; CO formed due to incomplete combustion of carbon forms carboxyhaemoglobin decreasing oxygen content of blood/impair visual activity

Or

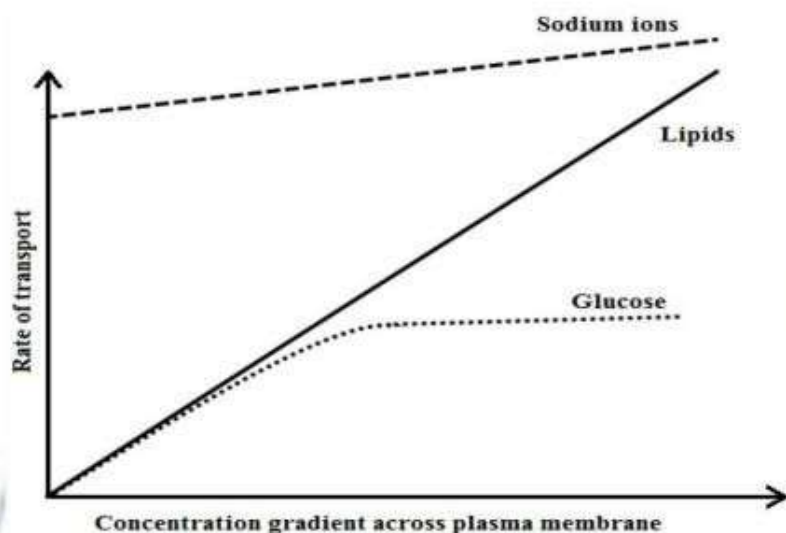
Because the conc of ozone at low altitude is boosted by the action of sunlight making it chemically very reactive, irritating eyes and respiratory tissues.



(vi) **Explain why the peak of the Hydrocarbons is higher than any other pollutant.**

This is because of the combustion of the Diesel/petrol produces several types of the volatile hydrocarbons eg benzene/ethene/butene compared while other pollutants are less in variety/form.

13. **The graph below shows the rates of transporting sodium ions, lipids and glucose across the membrane of living cells.**



(a) **Describe the rate of transport of each substance across the plasma membrane.**

- At low concentration gradient, the rate of sodium ions transport is high across the plasma membrane. Increase in concentration gradient across plasma membrane, the rate of transport of sodium ions increases gradually.
- At low concentration gradient, the rate of transport of lipids is low across plasma membrane. Increase in concentration gradient, the rate of transport of lipids increases rapidly across plasma membrane.
- At low concentration gradient, the rate of transport of glucose is low across plasma membrane is low across plasma membrane.
- Increase in concentration gradient, the rate of transport of glucose increases gradually and then becomes relatively constant.

(b) **Account for the observed trends in the transport of sodium ions and glucose.**

#### **Sodium ions**

At low concentration gradient, the rate of transport of sodium ions is high across the plasma membrane; increase in concentration gradient, the rate of transport of sodium ions increases gradually across the plasma membrane. This is because sodium ions are actively pumped across plasma membrane by the use of ATP and the concentration gradient has no effect on them since transport is against the concentration gradient.

#### **Glucose**

Increase in concentration gradient across plasma membrane, the rate of transport of glucose increases gradually. This is because of many channel proteins that are available to transport molecules.

**(c) Compare the rate of lipid and glucose transport.**

**Similarities**

Both the rate of transport of glucose and lipids are low across the plasma membrane at low concentration gradient. Both the rate of transport of glucose and lipids increases rapidly across the plasma membrane with increase in concentration gradient.

**Differences**

Further increase in concentration gradient across plasma membrane is the rate of transport of glucose is constant while that of lipids increases rapidly.

**(d) Explain how the mechanisms of transportation of substances in (b) above are beneficial and disadvantageous to cells**

**Advantages;**

It allows movement of large polar and charged molecules e.g. amino acids across the plasma membrane without expenditure of energy.

**Disadvantages**

Materials cannot be acquired against their concentration gradient hence cannot be acquired in very large quantities. The channel proteins can easily become saturated hence limiting the transportation of materials.

**Active transport**

Materials can be transported against their concentration gradients

**Disadvantages**

There is expenditure of energy in form of ATP.

**(ii) Outline the adaptations cell membranes for the mechanisms of transport explained in (d) (i) above.**

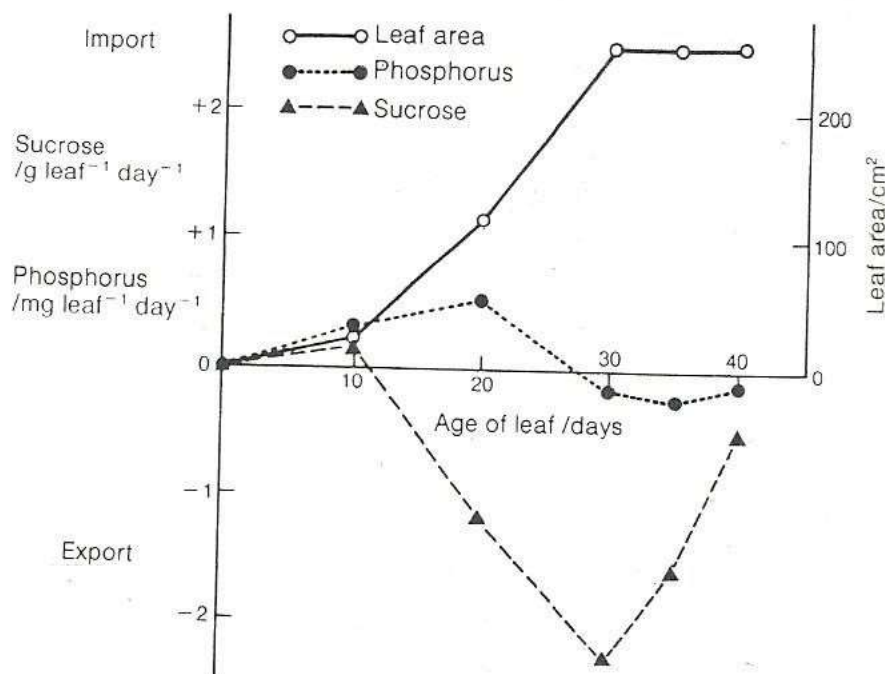
**For facilitated diffusion**

Cell membranes have special proteins molecules which are large and span the lipid bilayer; these are two types; channel proteins have fixed shape but with hydrophilic pores for transport of large polar molecules and ions; carrier proteins rapidly change shape to bind specifically to the molecules they assist to carry over; some membranes are highly folded forming microvilli; these increase the surface area for insertion of channel and carrier proteins;

**For active transport.**

Presence of carrier proteins on to which the phosphate binds for active transportation of materials; some membranes are highly folded forming microvilli; these increase the surface area for insertion of carrier proteins;

14. Figure 1 below shows the growth of a single plant leaf with its **import (+)** and **export (-)** of sucrose and phosphorus during a period of 40 days. Study the figure and answer the questions that follow.



(a) **From figure 1 above, explain the relationship between:**

**i) Leaf area and movement of sucrose.**

As leaf area increases, import of source into the leaf because sucrose is needed by the leaf as a respiratory substrate which is broken down to produce energy in form of Adenosine Tri Phosphate, required for cell division as the leaf is still undergoing growth;

As leaf area increases, export of sucrose out of the leaf increases because the rate of photosynthesis increases with an increase in leaf area; thus, more sugars are produced in the leaf, including sucrose a form in which sugars are transported from the leaf to storage organs; thus, increasing export of sucrose from the leaf;

As leaf area remains constant, the export of sucrose out of the leaf decreases because the rate of photosynthesis decreases as the leaf is aging; thus, less sugars are produced less sucrose is exported from the leaf;

**ii) Age of leaf and movement of phosphorus.**

As the age of the leaf increases, the import of phosphorus increases because the phosphorus is required for synthesis of Adenosine Tri Phosphate for cell division to occur so that leaf size increases the phosphorus is also needed for manufacture of some proteins and phospholipids in membranes;

As the age of the leaf increases, the import of phosphorus decreases because less proteins are made by the leaf since the leaf is aging.

As the age of the leaf increases, the export of phosphorus out of the leaf increases because phosphorus is exported out of the old leaf in form of phosphate into phloem sieve tubes to younger leaves so as to withdraw nutrients from the old leaf before it falls off the plant.

(b) Describe how the following are imported and exported through the leaf:

iii) Sucrose.

**Imported into the leaf.**

Glucose formed during photosynthesis combines with fructose, to form sucrose, that can be used as a respiratory substrate and a form in which carbohydrate is transported in plants.

**Exported from the leaf**

Accumulation of sucrose in the photosynthesizing cells reduces their water potential; sucrose is actively pumped into the phloem sieve cells of the leaf water which has been transported up to the stem xylem enters these mesophyll cells by osmosis due to accumulation of sucrose this causes an increase in the pressure potential of the leaf cells including the leaf sieve tube elements more than that of the cells in the sink that is the mesophyll cells

The food solution in the sieve tubes moves from a region of higher pressure potential in the leaf into that of lower pressure potential in the sink such as roots.

iv) Phosphorus.

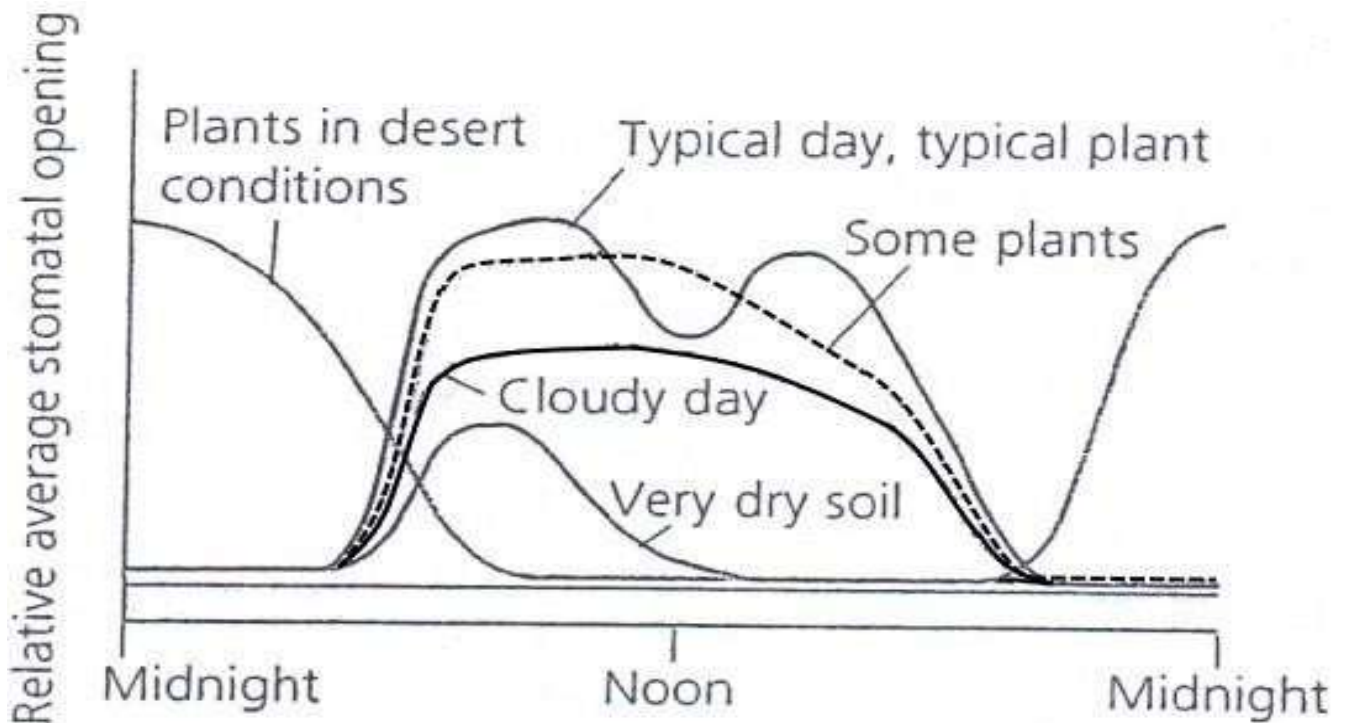
**Imported into the leaf.**

Phosphorus is imported in form of phosphate ions by active transport mainly by root hairs the phosphate ions then move by diffusion across the cortex and their controlled transfer across the endodermis the ions are actively pumped into the roots xylem by transfer cells of the pericycle in the xylem there is passive ascent up the xylem by transpiration stream the ions are then actively transported into leaf mesophyll cells by transfer cells of the leaf veins.

**Exported from the leaf**

Phosphorus is exported from an aging leaf in form of phosphate ions into phloem sieve tubes that transport the ions to younger organs of the plant.

Figure 2 shows how external factors affect stomatal opening in various plants. Study the figure and answer the questions that follow.



**Fig. 2**

- (c) Compared with a typical plant on a typical day, describe the effect of the following conditions on stomatal movement:
- (i) Very dry soil.
- Towards noon, very dry soil leads to an increase in the stomatal opening of atypical plant soil leads to typical day.
  - After noon, very dry soil causes an increase in the stomatal opening of a typical plant on a typical day
  - Very dry soil leads to a decrease in the stomatal opening of a typical plant on a typical day.
  - Cloudy day leads to an increase in the stomatal opening of a typical plant on a typical day.
  - Towards noon, cloudy day leads to a decrease in the stomatal opening of a typical plant.

**d ) Considering the plants in desert conditions;**

**(i) What are the advantages and disadvantages of the observed stomatal movement?**

From the figure, stomata of plants in desert open at night and close during day; the advantage of this is that water is not lost by the plant during day when the temperatures are too high and the rate of transpiration is also high; so the plant does not dry out; the disadvantage is that during day when stomata are closed, the plant cannot take up carbon dioxide; so probably no food is made by photosynthesis; additionally, there is little water taken up by the plant during day as the transpiration pull is reduced.

**ii) Explain how structural features are indicative of their habitat.**

- Waxy cuticle impermeable to water and thus reduce water loss by transpiration.
- Reduction in leaf size presenting a small surface area for reducing transpiration rate.
- Reduced stomatal number; to reduce transpiration rate
- Sunken stomata to trap air around stomata reducing transpiration rate
- Hairy leaves; trap air around leaves and reduce transpiration rate.
- Deep extensive root system; for increased water uptake from the water table.
- Scaly leaves; to reduce water loss by transpiration

**15 (a) Outline the features which are unique to the skeletal muscle.**

- it is striated
- It is multinucleate
- Muscle fibres are cylindrical
- It is voluntary
- It is prone to fatigue

**(b) How is the structure of myosin filament suited to its role in the muscle contraction?**

- Made up of thick fibrous protein for extra strength
- Has myosin heads which attach on the ADP sites of actin to form cross bridges
- Myosin heads have ATP which releases energy required for sliding of actin over myosin
- Stimulation of sarcolemma

**(c) Describe the events which lead to the shortening of the skeletal muscle.**

Stimulation of sarcolemma; current flows down to the T-tubes; action potential travels to the sarcoplasmic reticulum causing the release of  $\text{Ca}^{2+}$ ; the  $\text{Ca}^{2+}$  binds to the troponin causing a change of shape of tropomyosin and in the process exposing the ADP sites on actin filament; myosin heads bond with actin; ATP breaks down and the energy released causes the myosin head to swivel. This causes actin filament to slide past myosin. The muscle shortens/contracts;

**(d) Explain why rigor mortis occurs after death.**

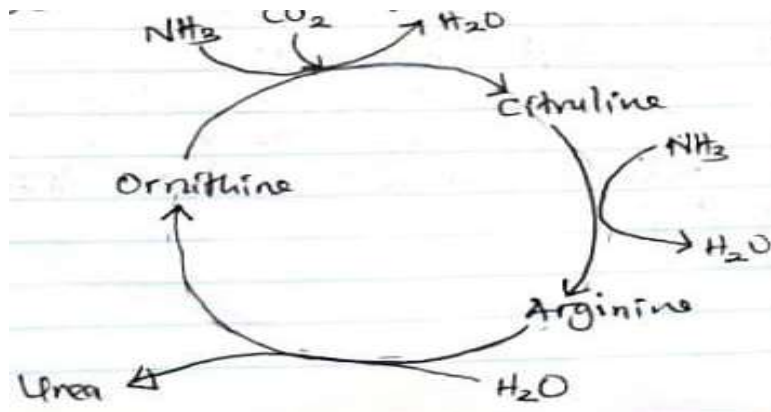
After death muscles contract and become rigid even without action potential; it is caused from loss of all ATP; which is required to cause separation of cross bridges from the actin filament.

**16 (a) Explain why nitrogenous waste do not normally occur in plants.**

Nitrogenous excretory products result from the breakdown of excess amino acids consumed by animals; in the case of plants, they can make all their amino acids from basic raw materials ( $\text{CO}_2$ ,  $\text{H}_2\text{O}$  and  $\text{NO}_2$ ); they only manufacture them as and when the need arises, they therefore have no excess amino acids to breakdown and hence no nitrogen waste.

**b) How does the human body deal with excess amino acids.**

- The excess amino acids are deaminated by the hepatocytes in the liver. The amino group and hydrogen atom are removed from the amino acid molecule to form ammonia. The amino acid residue is converted to either lipid or carbohydrate for storage or metabolized to generate energy.
- The ammonia formed is highly toxic. It is converted into less toxic urea in the ornithine cycle.
- A molecule of ammonia, carbon dioxide and Ornithine combine to form water and citrulline. A second ammonia molecule combines with citrulline to form arginine and a water molecule. Arginine combines with a water molecule to form urea and from ammonia consumes energy in form of ATP.



c) In the mammalian kidneys the relative length of the loops of Henle shows considerable variation from one species to another. Suggest with reasons type of habitats in which you expect to find species with extremely long and short loops of Henle.

- The longer the loops of the Henle the greater the proportion of water that is reabsorbed. Long loops of Henle are common in animals which need to conserve water e.g., animals living in hot deserts. On the other hand, the shorter the loops of Henle the lower the proportion of water that is reabsorbed. Short loops of Henle are common in animals which need to excrete a lot of water in their urine e.g., fresh water fish.

- (d) **Nitrogenous waste in animals may occur as ammonia, urea or uric acid. The table below shows the percentage of these three compounds in the urine of four different animals.**

|                  | <b>Ammonia</b> | <b>Urea</b> | <b>Uric acid</b> |
|------------------|----------------|-------------|------------------|
| Fresh water fish | 56             | 6           | 0                |
| Sea water fish   | 7              | 81          | 0                |
| Lizard           | 0              | 0           | 91               |
| Bird             | 3              | 4           | 72               |

- (i) **Explain the difference in the main excretory compound in fresh and sea water fish.**

While ammonia is the simplest excretory product it is very toxic; however, fresh water fish are able to dilute this ammonia to harmless concentrations using a large volume of water which passes into them osmotically and has to be removed by the kidneys; as sea water fish tend to lose water osmotically; they cannot dilute ammonia adequately

- (ii) **Both the lizards and birds are terrestrial and egg-laying. How do these characteristics relate to the nature of their main excretory products?**

As eggs are enclosed by shells excretory products cannot escape; if these were soluble, they would diffuse throughout the eggs and poison the embryo; the insoluble uric acid is therefore the only safe form in which excretory products can be stored

**a) Explain why it is important to study food webs rather than food chains in an eco-system**

- Unlike food chains, food webs are more realistic as most organisms have many predators and also prey too many other organisms.
- Food webs reveal more trophic levels than food chains
- Food webs stabilize the ecosystems and show more energy transfer unlike food chains

**b) Consider the trophic levels of pyramid of numbers and illustrate how energy is lost in passing through the levels**

- Primary source of energy is the sun, (95%) of the energy is lost through radiations or reflections; little energy (5%) of the energy is absorbed by the green plants or other photosynthetic organisms. Producers manufacture organic matter by photosynthesis and stored as (gross productivity).
- Some energy is used by plants in respiration or lost in form of fruit fall and leaf fall; the remaining energy is stored as net productivity. Organic matter in plants is consumed by the primary consumers mainly herbivores. Primary consumers are then eaten by secondary consumers which are also fed on by tertiary consumers. At each trophic level, certain proportion of energy is lost through egestion, excretion and respiration. Decomposition utilizes energy lost through egestion, excretion and dead organic matter then nutrients finally get recycled into the environment.



**c) Describe the ways in which nitrogen is incorporated into food chain**

- Nitrogenous waste products such as urea, proteins are contained in dead organic matters are putrefied to ammonia by putrefying bacteria. Through chemical combination of ammonia is further converted to ammonium compounds.
- With the aid of nitrifying bacteria such as nitrosamines ammonium compounds got oxidized to nitrites, which also get oxidized to nitrates by nitrobacteria. A nitrate gets absorbed by the plants, nitrogen is then incorporated into plant proteins enters the food chain.
- Through lightening, where nitrogen reacts with oxygen to form nitrogen dioxide which form nitrate and can be absorbed by the plant roots.
- Nitrogen fixation by nitrogen fixing bacteria e.g. rhizobium in root nodules of the leguminous plants.
- Industrial Haber process produces ammonia that is released into the soil.
- Application of organic fertilizers like CAN
- Sewage discharge on land since sewage contains ammonium compounds.

**END**